

Model Card - Census

Model Details

- Developed by Saba Ahmadi and Mahsa Najimoghaddam (equal collaboration, names are sorted alphabetically), 2021, V1., for IFT6390 course homework at the University of Montreal.
- Logistic regression (For specific detail on the model please read the paper.pdf in the same directory of this file).
- Trained on the census data (download from [UCI website](#) for binary classification of people who receives more income than 50K/year or less than or equal to 50K/year.
- If you have any question please send an email with subject[Census-ModelCard] to saba.ahmadi96@gmail.com or naji.mahsa@gmail.com.

Intended Use and Out of the Scope Use Cases

- Classifying people who receives more income than 50K or less than can be used for calculating the risk that a person have sufficient income to pay a loan back or not.
- Can be used in tax system to estimate the range of a person income, to monitor if people are lying about their income range.
- Can be used for analysing the gap of outcome ranges for people in minority and who are vulnerable to discrimination.
- Should not be used for people outside of the United States as it is trained on data of the United States.
- Should not be used in companies for estimating the offer of salary they should give to a person.

Factors

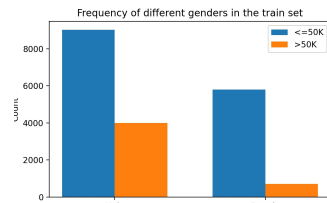
- The data set contains people who can be objective to lower income such as women, other races than white people. We calculate our model performance for different sex and different races and their intersection for instance black women.

Metrics

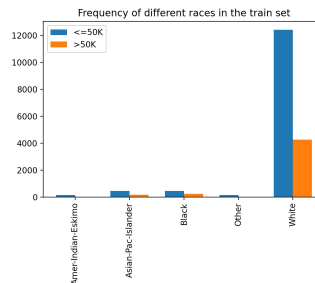
- We report False Positive Rate (incorrectly classified as high income receiver), False Negative Rate (incorrectly classified as low income receiver) , True Positive Rate(correctly classified as high income receiver) and True Negative Rate(correctly classified as low income receiver). As our train data is highly imbalanced this metrics can help to measure model performance errors across subgroups, and indicate the fairness and discrimination among them. (positive = '>50K', negative = '<=50K')
- All metrics reported at the .5 decision threshold.

Data

- Data is highly imbalanced. We provided the graphs to show the imbalance for different sex and races for the train set. We split our data randomly as follows: 60% of total data for training 20% for validation and 20% for final testing.



Frequency by sex



Frequency by race

Ethical Considerations

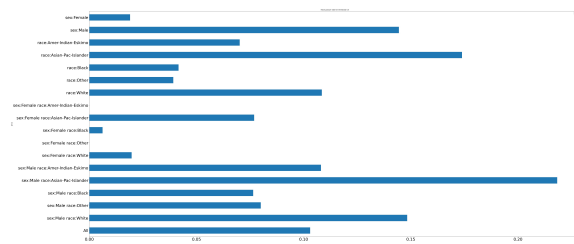
- The data does not consider people from any other genders than Male or Female.
- If this model will be used for identifying if a person receives high income for paying tax, it will be unfair as it has moderately high false positive rate for people in minority such as intersection of male and american-indian-eskimo.

Caveats and Recommendations

- We assumed only the features we have has impact on predicting the the outcome range, however; in reality there are some other factors that should be considered, for instance illegal outcome is not reported anywhere such as drug dealers outcome or what people may earned from buying digital currency. As people reported outcome or job may not be accurate our model is highly vulnerable to classify people with high salaries as low-income receiver.
- However, the model is trained in 2021, the data is out-dated and is for more than 20 years ago. So, this trained model cannot be used accurately for the released date as measures like inflammation is not considered at all.
- Our recommendation for future work is gathering exact data of salaries to change this problem to regression problem. Also, considering genders other than just male and female can help for more analysis of potential discrimination against genders in minority.

Quantitative Analyses

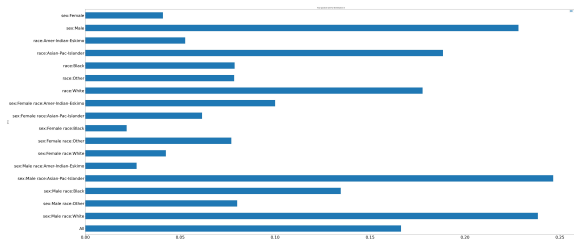
- You can see our model performance to compare it for different subgroups.



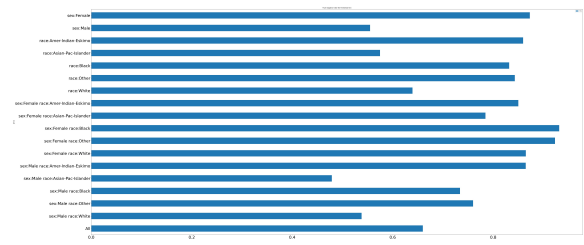
False positive rate



False negative rate



Reporting True positive rate



True negative rate

It is noteworthy that we used [overleaf model card template](#) for this latex file.